

# Prob 1 - Numerical sol. of ODEs

$$y' = (1 - 2t)y, \quad y(0) = 1$$

$$y(t) = e^{t-t^2}$$

a)  $h = 0.5$

$$y_{n+1} = y_n + h f(t_n, y_n) = y_n + h y'(t_n, y_n)$$

$$y_0 = 1$$

$$y_1 = 1 + 0.5(1 - 2 \cdot 0.5) \cdot 1 = 1$$

$$y_2 = 1 + 0.5(1 - 2 \cdot 1) \cdot 1 = 0.5 = y(1)$$

b)  $y_{n+1} = y_n + h y'(t_n, y_n) + \frac{h^2}{2} y''(t_n, y_n)$

$$y'' = -2y$$

$$y_0 = 1$$

$$y_1 = 1 + 0.5(1 - 2 \cdot 0.5) \cdot 1 + \frac{0.5^2}{2} \cdot (-2 \cdot 1)$$

$$= 1 + 0 + 0.125 = 1.125$$

$$y_2 = 1.125 + 0.5(1.125 - 2 \cdot 1) \cdot 1.125 + \frac{0.5^2}{2} \cdot (-2 \cdot 1.125)$$

$$= 1.125 - 0.5625 - 0.3125$$

$$= 0.25$$